

MoneyTracker

Complete App Documentation

Privacy-first · Fully local · No internet permission
SQLCipher encrypted · Android / Kotlin · V1 → V5 + UX Layer

FEATURE CATEGORIES

CORE	DESIGN & UX	SECURITY	ANALYTICS
DATA	NOTIFICATIONS	AI	PRIVACY

V1 Foundation	V2 Budgeting	V+ UX Layer	V3 Insights	V4 On-Device AI	V5 Polished
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Complete Feature Summary	All features — categories, versions, and permissions

Introduction

What is MoneyTracker?

MoneyTracker is a fully local, privacy-first Android expense tracking application built with Kotlin and Android Studio. It stores every piece of financial data on-device inside a SQLCipher-encrypted Room Database, requires zero internet permissions, and is architecturally incapable of transmitting data to any server. Released on Google Play as a free application with no advertising and no monetisation plans.

Core Privacy Philosophy

Privacy is not a feature — it is the foundation. Every architecture decision flows from one question: does this require data to leave the device? If yes, it is not built. The absence of the INTERNET permission in AndroidManifest.xml is the proof, not a promise. Users can verify this in Android Settings or by decompiling the APK. This guarantee holds across all versions.

Technology Stack

Language: Kotlin. UI: Jetpack Compose with Material You (Material Design 3). Database: Room encrypted with SQLCipher AES-256. Charts: MPAndroidChart. Background work: WorkManager. AI (V4 only): MediaPipe LLM Inference API with Gemma on-device model. No cloud SDKs, no analytics libraries, no advertising SDKs included anywhere.

Roadmap Overview

Six phases: V1 (core tracker) → V2 (budgeting) → V♦ UX Layer (experience enhancements from user research) → V3 (advanced insights) → V4 (on-device AI) → V5 (polished Play Store release). Each phase is a shippable complete product — not a work in progress.

Development Timeline

At one hour of focused daily development: V1 month 8, V2 month 12, V♦ month 14, V3 month 17, V4 month 21, V5 month 25. These are guides, not deadlines — learning and code quality take priority. Path: Kotlin Koans → Udacity ud9012 → Google Android Kotlin Fundamentals codelabs → practice notes app → V1 development.

Design Principles

Every screen must answer the user's question in under two seconds. Every interaction must feel intentional and polished. No feature should require the user to understand its internals — only what it does for them. Consistency across all screens builds trust. The app should feel designed by someone who deeply respects the user's time and privacy.

V1

Foundation

The essential core: track every rupee, encrypt everything, require nothing from the internet.

V1 is the bedrock. Intentionally minimal — every feature is necessary, nothing unnecessary. The security architecture built here underpins all future versions. V1 is a complete shippable product.

CORE

Expense & Income Entry

Introduced in V1

◆ **UX INSIGHT** · Top-rated apps show users spend just 1–3 minutes daily on transactions. Entry must complete in under 10 seconds. Every extra tap costs long-term consistency.

WHAT IT DOES

Tap a floating action button to open the entry screen: amount (purpose-built numeric keypad), type (expense/income), category, date (defaults today), optional note to 200 chars. Committed to the encrypted Room Database on confirmation.

WHY IT MATTERS

Every insight flows from accurate data. A fast, friction-free entry experience is the single most important factor in building the daily logging habit.

TECHNICAL DETAILS

- Dedicated numeric keypad — not the system keyboard
- Supports decimal amounts to two places (e.g. ■245.50)
- Date picker allows backdating for forgotten transactions
- Edit any past transaction by tapping it in the list
- Swipe-to-delete with undo snackbar to prevent accidents
- Swipe-right gesture to quickly duplicate a recent transaction

CORE

Custom Categories

Introduced in V1

◆ **UX INSIGHT** · Emoji labels and colour-coded categories significantly improve engagement. Users who personalise categories feel ownership and log more consistently.

WHAT IT DOES

Users create, rename, reorder, and delete categories. Each has a name, a colour from a 20-colour palette, and an optional emoji icon. Sensible defaults pre-loaded on first launch (Food, Transport, Bills, Shopping, Health, Entertainment, Savings, Other).

WHY IT MATTERS

Categories are personal. Forcing users into a fixed taxonomy means the app tracks someone else's spending. Full customisation makes the data meaningful.

TECHNICAL DETAILS

- Up to 30 categories simultaneously
- 20-colour palette + emoji picker for each category
- Drag-to-reorder in the category management screen
- Categories with transactions cannot be deleted — only archived
- Archived categories hidden from entry but preserved in history

SECURITY

SQLCipher-Encrypted Room Database

Introduced in V1

WHAT IT DOES

All data stored in Room Database encrypted with SQLCipher AES-256-CBC. Every database page is encrypted at rest. Key derived from user's PIN using PBKDF2 (100,000 iterations, SHA-256). Without the correct PIN, the database file is indistinguishable from random bytes.

WHY IT MATTERS

Financial data is among the most sensitive on a phone. If the device is lost or forensically examined, encryption ensures data cannot be read without the PIN — even with direct storage access.

TECHNICAL DETAILS

- SQLCipher: open-source, independently audited, widely deployed
- AES-256-CBC: encryption per database page, not per field
- PBKDF2: 100,000 iterations, SHA-256, random salt
- Salt stored separately from the database file
- PIN change triggers key re-derivation and database re-encryption
- Encryption cannot be disabled — no plaintext fallback

PRIVACY

No Internet Permission

Introduced in V1

WHAT IT DOES

The AndroidManifest.xml contains no INTERNET permission. Android enforces this at OS level — the app cannot open any network socket regardless of what code attempts. This is an OS-guaranteed, irrevocable constraint applying to all library code bundled in the APK.

WHY IT MATTERS

Any app can claim it doesn't send data. Only an app without INTERNET can prove it. This is MoneyTracker's most fundamental privacy guarantee and will never be added in any version.

TECHNICAL DETAILS

- Verifiable: Settings > Apps > MoneyTracker > Permissions
- Also verifiable by decompiling the APK and inspecting AndroidManifest.xml
- Applies to SQLCipher, MPAndroidChart, and every bundled library
- V3 adds READ_SMS for bank SMS parsing — INTERNET is never added

DESIGN & UX

Monthly Dashboard

Introduced in V1

◆ **UX INSIGHT** · Top-rated finance apps surface ONE metric as the hero element. For MoneyTracker: the net balance — large, centred, colour-coded. Everything else is secondary.

WHAT IT DOES

Home screen shows current month totals at a glance: total income, total expenses, and net balance as the hero element. Below: scrollable recent transactions with category icon, amount, and date. Month selector navigates to any past month.

WHY IT MATTERS

The dashboard is the screen users see most. It must answer 'How am I doing this month?' in under two seconds. Clarity here is what builds the daily habit.

TECHNICAL DETAILS

- Net balance shown large and centred — green (positive) or red (negative)
- Month selector: swipe left/right or tap the month label
- Tap any transaction to open full detail and edit screen
- Animated summary cards on load for visual polish
- Budget health compact indicator below the summary cards

ANALYTICS

Reports — Charts

Introduced in V1

◆ **UX INSIGHT** · Users of Copilot and Monarch describe charts as 'life-changing' for self-awareness. Beautiful, animated charts are the reward for logging consistently.

WHAT IT DOES

Reports screen renders pie and bar charts using MPAndroidChart. Pie chart shows each category's share of total spending. Bar chart shows daily spending totals across the month. Both navigable by month.

WHY IT MATTERS

Numbers in a list tell you what happened. Charts show the shape of spending — which category dominates, whether spending spikes mid-month, which days cost most.

TECHNICAL DETAILS

- Pie chart slices match each category's assigned colour
- Tap a pie slice to see exact amount and percentage
- Bar chart colour-coded by transaction type
- Charts load with smooth animations via MPAndroidChart animators
- All charts rendered entirely from local Room Database data

DESIGN & UX

Dark & Light Theme

Introduced in V1

WHAT IT DOES

Full Material You (Material Design 3) theming with dark and light modes. Respects system-level dark mode setting by default, with manual override in Settings. All colour tokens, charts, and category colours adapt consistently across both themes.

WHY IT MATTERS

Dark mode reduces eye strain in low light, is preferred for late-night use, and reduces battery drain on OLED panels — standard on most modern Android phones.

TECHNICAL DETAILS

- Follows system setting by default; manual override in Settings
- Custom category colours adapt automatically to both themes
- MPAndroidChart themes re-applied on change without flickering

◆ **UX INSIGHT** · Biometric unlock must feel instant — under half a second. A slow lock screen damages perception of quality more than almost any other UX failure.

WHAT IT DOES

On open (and after configurable inactivity timeout), MoneyTracker requires authentication via PIN or device biometrics using Android's BiometricPrompt API. Second, independent security layer on top of database encryption.

WHY IT MATTERS

Encryption protects data when device is off. App lock protects it when device is on but unlocked — if someone picks up an unlocked phone. Both layers are needed.

TECHNICAL DETAILS

- Configurable timeout: immediate, 30 s, 1 min, 5 min
- BiometricPrompt API: fingerprint and face unlock
- PIN fallback always available if biometrics fail
- Lock activates on background after timeout expires

V2

Smart Budgeting

Plan spending, set goals, get reminded, and export your data — all without internet.

V2 transforms MoneyTracker from a passive ledger into an active budgeting tool. Users can plan their month in advance, set financial goals with visible progress, receive timely local reminders for bills, view spending on a calendar, and export their full transaction history. Every feature in V2 is fully offline.

CORE

Category Budgets

Introduced in V2

◆ **UX INSIGHT** · Progress bars are the most praised visual element in every top-rated finance app. Never block spending when a budget is exceeded — that creates distrust.

WHAT IT DOES

Users assign a monthly spending limit to any category. Budget screen shows a live progress bar per budgeted category that fills as spending accumulates. Bar turns amber at 80%, red when exceeded. Dashboard surfaces a compact budget health summary.

WHY IT MATTERS

Knowing what you spent is informative. Knowing you are about to overspend is actionable. Budgets shift MoneyTracker from a retrospective diary into a forward-looking tool.

TECHNICAL DETAILS

- Set budgets per category independently — not all categories need one
- Resets on the 1st of each month
- Progress: green (under 80%) → amber (80–99%) → red (100%+)
- Optional local notification at 80% budget usage
- Budget vs. actual comparison chart in reports
- Budget never blocks a transaction — informational only

CORE

Rollover Budgets

Introduced in V2

WHAT IT DOES

When enabled, any unspent portion of a month's budget carries forward to the next month. Example: Food budget ■5,000, spent ■3,800 — next month Food becomes ■6,200. Toggleable per category.

WHY IT MATTERS

Fixed budgets resetting to zero punish frugality. Rollover budgets make underspending feel valuable — you bank savings for months when you genuinely need more flexibility.

TECHNICAL DETAILS

- Toggled independently per category
- Accumulated rollover shown separately from base budget
- Optional cap: maximum rollover limit to prevent unlimited accumulation
- Rollover history viewable per category

CORE

Savings Goals

Introduced in V2

◆ **UX INSIGHT** · Goal completion animations are among the most positively mentioned UX moments in finance app reviews. Archive completed goals — never delete. The history of past wins motivates future ones.

WHAT IT DOES

Users create named goals (e.g. 'New Laptop — █40,000') with a target amount and optional date. Contributions added manually. Each goal shows a progress bar, amount saved, remaining, and an estimated completion date based on trailing 3-month average contribution rate.

WHY IT MATTERS

Vague intentions to 'save more' fail. Named goals with visible progress make saving concrete and motivating. The progress bar creates a feedback loop encouraging consistent contributions.

TECHNICAL DETAILS

- Unlimited simultaneous goals
- Contributions manual — user decides when to allocate money
- Completion estimate: trailing 3-month average contribution rate
- Goal states: Active, Paused, Reached (archived with celebration, never deleted)
- Optional nudge: 'You haven't contributed to [goal] in 2 weeks'

NOTIFICATIONS

Bill & Recurring Expense Reminders

Introduced in V2

◆ **UX INSIGHT** · Users praise configurable lead time — reminded 3 days before a bill, not on the day. The key is giving enough time to actually act on the information.

WHAT IT DOES

Users log recurring bills with name, amount, due date, and recurrence pattern (monthly, weekly, custom). WorkManager schedules local Android notifications N days before each due date (configurable: 1–7 days). All notifications handled by the Android OS — no internet.

WHY IT MATTERS

Late fees are usually caused by forgetfulness, not inability to pay. A timely local reminder eliminates this entirely and helps users plan cash flow in advance.

TECHNICAL DETAILS

- WorkManager: notifications fire reliably even after device restart
- Reminder lead time configurable per bill (1–7 days prior)
- Recurring bills auto-reschedule after each due date passes
- Bills listed chronologically on an upcoming bills screen
- Marking a bill paid auto-logs it as a transaction (optional)
- Committed monthly total: sum of all active bills' monthly equivalents

ANALYTICS

Calendar View

Introduced in V2

WHAT IT DOES

Monthly calendar grid where each date cell shows that day's total spending. Higher-spend dates tinted more intensely (heat-map effect). Tapping a date expands a panel showing all transactions for that day.

WHY IT MATTERS

Lists and charts show aggregates. The calendar reveals timing — whether spending clusters on weekends, at month-end, or specific recurring dates. This temporal dimension is invisible in other views.

TECHNICAL DETAILS

- Heat-map colouring: deeper teal = higher spending
- Today highlighted with an accent ring
- Tapped day: full transaction list + income/expense totals
- Income and expense totals shown side by side per day

DATA

CSV Data Export

Introduced in V2

WHAT IT DOES

Export full transaction history — or a filtered subset — as a standard CSV file saved to a user-chosen local folder via Android's Storage Access Framework. Columns: Date, Amount, Currency, Type, Category, Note. The file never leaves the device.

WHY IT MATTERS

Data portability is a privacy right. Users should extract their own data in a standard format for backup, spreadsheet analysis, or sharing with a financial advisor — on their terms.

TECHNICAL DETAILS

- Date range filter: all time, this month, last 3 months, or custom
- Category filter: single or all
- File naming: MoneyTracker_export_YYYY-MM-DD.csv
- Saved via SAF — user chooses destination folder
- Amounts always positive; Type column distinguishes Expense/Income

CORE

Transaction Search & Filter

Introduced in V2

WHAT IT DOES

Search bar matches keyword input against amount, category name, and note text. Filter chips narrow by date range, type (expense/income), and category. Results update in real time via Room Database Full Text Search (FTS).

WHY IT MATTERS

After months of data, scrolling to find a transaction is impractical. Search and filter turn an accumulating list into an instantly queryable personal financial record.

TECHNICAL DETAILS

- Real-time results via Room FTS as the user types
- Filter chips: date range, type, category — all combinable
- Result count shown in toolbar
- Recent searches saved locally for quick re-use

◆ **UX INSIGHT** · Across all top finance apps the home screen widget is the strongest passive habit-builder. Users check their balance multiple times daily from the widget alone — without opening the app.

WHAT IT DOES

Resizable Android home screen widget showing current month: total spent, total income, and net balance. Updates every time a transaction is logged. Tapping opens MoneyTracker.

WHY IT MATTERS

Ambient awareness nudges better spending decisions passively throughout the day. Seeing financial status on the home screen builds habit without requiring willpower.

TECHNICAL DETAILS

- Small (2x1) and medium (4x2) sizes
- Updates on transaction log and at midnight reset
- Respects system dark/light mode
- Budget progress bar visible in medium widget size



UX Layer

Research-driven experience enhancements — the features users love but rarely think to ask for.

The UX Layer is a dedicated phase of improvements drawn from deep research into what users of top-rated finance apps (Copilot, Monarch, YNAB, PocketGuard) praise most in reviews. These are not new data features — they are refinements that make existing features dramatically more satisfying to use daily. All features are fully offline.

- All features in this phase come from deep research into what users of top-rated finance apps (Copilot, Monarch, YNAB, PocketGuard) praise most in reviews. None require internet. None change the privacy model.

DESIGN & UX

Safe-to-Spend Number

Introduced in V◆

◆ **UX INSIGHT** · PocketGuard built an entire product around this single number. Users report it becomes the one figure they check before any discretionary purchase — transforming abstract budget data into a single actionable permission.

WHAT IT DOES

A large 'Safe to Spend' number on the dashboard showing how much discretionary money remains this month. Calculated as: Income – Committed bills due this month – Already spent – Budgeted categories remaining. One number that answers 'Can I afford this?'

WHY IT MATTERS

Users don't want to do mental arithmetic across five categories. A single safe-to-spend number gives an immediate answer and reduces the cognitive load of every spending decision.

TECHNICAL DETAILS

- Displayed prominently below the net balance on the dashboard
- Updates instantly on every transaction logged
- Green (healthy), amber (low), red (overextended)
- Tapping shows the breakdown: income – bills – spent = safe to spend
- Fully derived from local data — no internet needed

DESIGN & UX

'Since You Last Opened' Summary

Introduced in V◆

◆ **UX INSIGHT** · Copilot's 'since last opened' summary is one of the most praised single UX decisions in any finance app. It makes every app open feel purposeful — there's always a clear starting point.

WHAT IT DOES

When the user opens MoneyTracker, if there are unreviewed transactions since their last session, the dashboard highlights them in a 'Since you were away' card at the top. A 'Mark all reviewed' button clears it.

WHY IT MATTERS

Opening to a static screen every day is passive. Opening to 'here's what happened since you last checked' makes every open feel active and keeps the user engaged.

TECHNICAL DETAILS

- Tracks last-opened timestamp locally
- Shows new transactions grouped by day since last open

- Unreviewed transaction count shown as badge on app icon
- 'Mark all reviewed' clears the card and resets the timestamp
- Individual transactions can be reviewed or edited from the card

DESIGN & UX

Pre-Committed Bills on Budget Screen

Introduced in V◆

◆ **UX INSIGHT** · Copilot pre-commits recurring charges at the start of each month so users never think they're under budget when a subscription is about to hit. One change that eliminates most end-of-month surprises.

WHAT IT DOES

Bills due this month from the Reminders feature are shown as 'Reserved' amounts on the budget screen — displayed as committed spend even before the due date. The progress bar accounts for these reserved amounts so users see their true available budget.

WHY IT MATTERS

A user with ■1,500 remaining who has a ■1,200 bill due tomorrow doesn't actually have ■1,500 to spend. Pre-committed bills make the true available budget visible.

TECHNICAL DETAILS

- Bills from Reminders feature automatically reserved on budget screen
- Reserved amounts shown in amber on the progress bar
- Tooltip: 'Includes ■X reserved for upcoming bills'
- When a bill is paid and logged, reserved converts to actual spent
- Toggle to show/hide reserved amounts for power users

CORE

Bulk Transaction Actions

Introduced in V◆

◆ **UX INSIGHT** · YNAB users specifically praise bulk transaction selection as a key efficiency feature. When importing SMS-parsed transactions in V3, bulk categorise becomes essential.

WHAT IT DOES

Long-press any transaction to enter multi-select mode. Tap additional transactions to add them to the selection. A contextual action bar appears at the bottom: Categorise all, Delete all, Export selection.

WHY IT MATTERS

After SMS auto-reading (V3) parses a batch, the user may need to re-categorise several at once. Doing this one-by-one is frustrating. Bulk actions make batch operations as fast as single ones.

TECHNICAL DETAILS

- Long-press activates multi-select mode
- Clear visual selection state on each tile
- Contextual bottom bar: Categorise, Delete, Export
- Categorise shows category picker and applies to all selected
- Selection count shown in toolbar
- Two-finger swipe selects a range of transactions

DESIGN & UX

Monthly Running Total in Transaction List

Introduced in V❖

❖ **UX INSIGHT** · Copilot users specifically call out 'monthly running totals' as very useful. It adds context to every transaction without requiring navigation to a separate screen.

WHAT IT DOES

A sticky section header above each day's transactions shows the cumulative month total at that point in time. As the user scrolls through the month they can see how the balance evolved day by day.

WHY IT MATTERS

A list of transactions is data. Seeing how the running total changed across the month reveals the story — which day caused a big jump, whether early or late-month spending is higher.

TECHNICAL DETAILS

- Running total shown in section header: 'April 10 · Month total: ■8,420'
- Green if under budget for that point in month, red if over
- Toggable in Settings for users who prefer a cleaner list view

NOTIFICATIONS

At-Risk Category Highlights on Dashboard

Introduced in V❖

❖ **UX INSIGHT** · Monarch and Copilot both surface at-risk budget categories prominently on the dashboard. Users praise this as the feature that most prevents overspending — because it's ambient.

WHAT IT DOES

The dashboard shows a compact 'Budget Alerts' row with categories that have crossed 80% of their monthly budget — category icon, percentage used, and amount remaining. Tapping opens the full budget screen.

WHY IT MATTERS

If budget warnings are buried in a sub-screen, users won't see them until after the problem. Surfacing at-risk categories on the home screen makes warnings impossible to miss.

TECHNICAL DETAILS

- Appears on dashboard only when ≥ 1 category is above 80%
- Maximum 3 categories shown — tap 'See all' for the rest
- Sorted by urgency: closest to limit shown first
- Amber for 80–99%, red for 100%+

DESIGN & UX

Swipe Gestures on Transaction Tiles

Introduced in V❖

❖ **UX INSIGHT** · Swipe gestures on transaction tiles are praised across all major finance apps as the interaction that makes daily maintenance feel satisfying. Small change, outsized impact on quality.

WHAT IT DOES

Swipe left to reveal a delete action with confirmation. Swipe right to reveal a quick-edit action opening the edit sheet inline. Both include haptic feedback.

WHY IT MATTERS

Tapping a transaction, navigating to detail, finding delete, and confirming is 4 steps. Swipe-and-confirm is 2. Across hundreds of transactions this efficiency compounds dramatically.

TECHNICAL DETAILS

- Swipe left: delete with inline undo snackbar (4 seconds)
- Swipe right: quick edit sheet — no full navigation
- Haptic feedback on reveal and action confirmation
- Requires deliberate swipe — not triggered by list scrolling
- Gesture hint shown on first open via subtle animated arrow

CORE

Subcategories

Introduced in V❖

❖ **UX INSIGHT** · Subcategories are among the most-requested features across every finance app review. Users want 'Food' as a parent but 'Groceries', 'Dining Out', 'Coffee' as children. Clean overview by default, granular detail on demand.

WHAT IT DOES

Each main category can have up to 10 subcategories. Subcategories inherit the parent's colour but have their own emoji and name. Reports show spending at parent level by default, expanding to subcategory level on tap. Budgets can be set at parent or subcategory level.

WHY IT MATTERS

Top-level categories like 'Food' are too broad for meaningful insight. But 50 top-level categories is overwhelming. Subcategories give both — clean overview, granular detail on demand.

TECHNICAL DETAILS

- Up to 10 subcategories per parent category
- Inherit parent colour; own emoji icon and name
- Transaction entry: select parent first, subcategory second (optional)
- Reports: parent totals default, expand to see subcategory breakdown
- Budgets: set at parent level (covers children) or per subcategory

V3

Advanced Insights

SMS auto-reading, cash flow projections, debt planning, net worth, multi-currency — all offline.

V3 adds the depth separating a basic tracker from a complete personal finance tool. SMS auto-reading reduces manual entry friction to near zero. Cash flow projections, debt payoff planner, manual net worth tracking, multi-currency, and spending pattern analysis give users an analytical layer that rivals paid apps — with no internet and no privacy compromise.

CORE

SMS Transaction Auto-Reading

Introduced in V3

WHAT IT DOES

With explicit READ_SMS permission, MoneyTracker monitors incoming SMS messages from recognised bank sender IDs. A regex pattern library extracts amount, merchant/description, and type from bank alert messages. App presents a pre-filled confirmation card — user must approve or dismiss before anything is saved. The raw SMS body is never stored.

WHY IT MATTERS

Manual entry is the biggest friction point. In India, virtually every bank transaction triggers an SMS alert. Auto-reading reduces logging from 10 seconds to 2 (just confirm), dramatically improving data completeness.

TECHNICAL DETAILS

- Requires READ_SMS — clearly explained before permission request
- Pattern library covers major Indian banks: SBI, HDFC, ICICI, Axis, Kotak, and more
- Pattern library extensible via app updates
- User reviews and edits pre-filled amount and category before confirming
- Raw SMS body never persisted — only extracted values
- Confirmation card can always be dismissed without saving

ANALYTICS

Cash Flow Projections

Introduced in V3

WHAT IT DOES

Using recurring bills and historical averages, MoneyTracker projects expected income, committed expenses, and discretionary spending for the next 1, 2, and 3 months. Shown as a waterfall chart with projected end-of-month balance. Pure rule-based arithmetic — no AI.

WHY IT MATTERS

Reactive tracking tells you what happened. Projections tell you what is likely to happen, enabling proactive decisions. Knowing month 2 looks tight lets the user adjust in month 1.

TECHNICAL DETAILS

- Layer 1: committed income — recurring income entries (high certainty)
- Layer 2: committed bills — all active recurring reminders (high certainty)
- Layer 3: discretionary — trailing 3-month category averages (lower certainty)
- Projection confidence shown as a visual range — not a false single number
- Tapping a projected month shows its itemised breakdown
- All calculations local — no external data sources

◆ **UX INSIGHT** · The delta between Snowball and Avalanche strategies is the single most valuable output. Make it prominent: 'Avalanche saves █8,400 and clears debt 3 months faster.' Concrete numbers transform an abstract concept into a motivating decision.

WHAT IT DOES

Users enter each debt: name, outstanding balance, annual interest rate, minimum monthly payment. Planner calculates total time and interest under Snowball (smallest balance first) and Avalanche (highest interest first). Both visualised as month-by-month payoff timelines.

WHY IT MATTERS

Debt is one of the most psychologically difficult financial situations. A clear plan showing a specific debt-free date turns an abstract burden into a solvable problem with a finish line.

TECHNICAL DETAILS

- █ Up to 10 debts simultaneously
- █ Extra monthly payment field: amount above minimums the user can contribute
- █ Side-by-side comparison: Snowball vs. Avalanche (total interest, total months)
- █ Month-by-month balance table per debt under chosen strategy
- █ Projected debt-free date shown prominently
- █ Interest saved by choosing Avalanche shown explicitly

WHAT IT DOES

Users manually log assets (savings, investments, devices, vehicles, property) and liabilities (outstanding loans, credit card dues, EMI's). Net worth = total assets – total liabilities, shown as a historical line graph growing over time as the user updates monthly.

WHY IT MATTERS

Daily expense tracking shows cash flow. Net worth shows wealth accumulation — the bigger picture. Watching net worth grow, even slowly, is one of the most motivating long-term metrics.

TECHNICAL DETAILS

- █ Assets: savings, investments, property, devices, vehicles, other
- █ Liabilities: loans, credit cards, EMI's, other debts
- █ Monthly snapshot model: user updates at end of each month
- █ Historical net worth graph from date of first entry
- █ Asset and liability breakdown in separate pie charts
- █ Entirely manual — no bank syncing, no internet

DATA

Multi-Currency Support

Introduced in V3

WHAT IT DOES

Transactions logged in any currency. Users define their base currency (e.g. INR) and add secondary currencies with manually set exchange rates. Foreign transactions stored with original currency and amount, converted to base for all aggregates and reports. Rates static until user updates them.

WHY IT MATTERS

Anyone who travels, shops internationally, or receives income in foreign currencies needs this. Since live rates require internet, MoneyTracker uses manual rates — keeping the user in control and the no-internet guarantee intact.

TECHNICAL DETAILS

- Base currency set during onboarding or in Settings
- Up to 10 secondary currencies configurable
- Manual exchange rate per currency (e.g. 1 USD = 83.50 INR)
- Exchange rate frozen at transaction time — historical data stays stable
- Transaction list shows original currency and base equivalent
- Rate updates do not retroactively change historical transactions

ANALYTICS

Spending Pattern Analysis (Rule-Based)

Introduced in V3

WHAT IT DOES

Analyses transaction history using deterministic statistical rules to surface automatically generated observations. Examples: 'Your weekend spending is 2.1x your weekday average,' 'Food spending up 22% over 3 months,' '0 on Entertainment in January — your lowest ever.'

WHY IT MATTERS

Raw data does not speak for itself. Surfacing patterns turns data into insight. This also lays the groundwork for V4's AI layer.

TECHNICAL DETAILS

- Insights refreshed weekly from local data
- Variance threshold: only surfaces if $|\text{variance \%}| > 15\%$
- Categories with under 2 months of data excluded
- Insight cards shown on dashboard and reports screen
- User can dismiss or save individual insights
- Maximum 5 insights shown at a time — ranked by magnitude

ANALYTICS

Advanced Reports

Introduced in V3

WHAT IT DOES

Expanded reporting beyond V1's monthly charts. Includes: year-over-year comparison (any two months side by side), 12-month category trend lines, day-of-week heatmap showing average spending by day of week, and a top-spending days list per month.

WHY IT MATTERS

Monthly charts show one month. Multi-month trends reveal whether financial behaviour is improving. Year-over-year comparison is absent from almost all simple expense trackers.

TECHNICAL DETAILS

- Year-over-year: any two months compared side by side
 - 12-month trend: line chart per category
 - Day-of-week heatmap: averaged over trailing 3 months
 - Top spending days list: 5 highest-spend days of any month
 - All charts rendered offline from local Room Database
-

V4

On-Device AI

Gemma runs entirely on your device — intelligent insights with zero data leaving your phone.

V4 integrates Google's MediaPipe LLM Inference API with the Gemma on-device language model. All AI inference runs on the device's NPU or GPU. No data ever sent to a cloud API. MoneyTracker's no-internet guarantee is fully preserved. V4 makes the app smarter without making it any less private.

AI

MediaPipe / Gemma On-Device Integration

Introduced in V4

WHAT IT DOES

MediaPipe LLM Inference Task integrated as an Android library. Gemma model weights downloaded once at app update time and stored in the app's private local storage. All inference runs on-device — no network call is ever made during AI feature use.

WHY IT MATTERS

Cloud AI requires sending financial data to a remote server — a fundamental privacy violation. On-device AI provides intelligent features without ever exposing private data. MediaPipe + Gemma is the technically correct choice for this constraint.

TECHNICAL DETAILS

- Model: Gemma (version selected based on device compatibility at V4 dev time)
- Inference API: MediaPipe LLM Inference Task for Android
- Model weights in app-private storage after initial download
- All subsequent use requires no download or network connection
- Loading indicator shown while model initialises on first use
- Inference triggered only when user explicitly uses an AI feature

AI

AI Spending Forecasts

Introduced in V4

WHAT IT DOES

Gemma analyses the user's full transaction history (provided as structured local context) and generates personalised monthly spending forecasts with natural language reasoning. Unlike V3's rule-based projections, Gemma identifies seasonal patterns and contextual anomalies invisible to simple arithmetic.

WHY IT MATTERS

Rule-based projections are accurate but rigid. Gemma produces richer, more nuanced predictions with articulated reasoning — making forecasts understandable and actionable.

TECHNICAL DETAILS

- Forecast horizon: 1 month and 3 months
- Output: category-by-category prediction + natural language summary
- User can add context ('I'm travelling next month') to refine
- Forecast history saved locally for comparison against actuals
- Confidence indicator shown per category forecast

AI

Natural Language Queries

Introduced in V4

WHAT IT DOES

Conversational query screen where users type plain-language questions about their finances. Gemma interprets the question, fetches relevant data from Room Database, and returns a natural language answer. Examples: 'What did I spend most on last quarter?' or 'How close am I to my laptop savings goal?'

WHY IT MATTERS

Not every user wants to navigate chart screens for a specific question. Conversational querying makes financial data accessible regardless of technical comfort level.

TECHNICAL DETAILS

- Query screen accessible from main navigation
- Gemma receives structured data summaries — not raw message content
- Follow-up questions supported within the same session
- Answers can include figures, comparisons, and trend observations
- Query history saved locally for reference

AI

Anomaly Detection

Introduced in V4

WHAT IT DOES

After each transaction is logged, Gemma compares it to the user's established patterns and flags entries that deviate significantly. Flagged entries show a subtle amber indicator. User can dismiss or add a note explaining the anomaly.

WHY IT MATTERS

Unusual transactions can indicate data entry errors, forgotten large purchases, or unexpected charges. Gentle flagging draws attention without being intrusive.

TECHNICAL DETAILS

- Runs after each transaction — not on every app open
- Threshold adapts to user's own data — not global averages
- Amber indicator on transaction tile (not an alarm)
- User can dismiss the flag or add an explanatory note
- Does not block or prevent saving any transaction

AI

Smart Category Suggestions

Introduced in V4

WHAT IT DOES

When a user types a merchant name or description during entry, Gemma suggests the most appropriate category based on text and the user's own history. Suggestion appears as a tappable chip — accept or ignore.

WHY IT MATTERS

Selecting a category from a list is minor friction that multiplies over hundreds of transactions. A confident one-tap suggestion reduces this to near zero.

TECHNICAL DETAILS

- Suggestion chip above the category picker with confidence percentage
- User can ignore and select manually — never forced
- Suggestions improve as categorisation history grows

- Merchant names never leave the device — fully local inference

AI

AI Savings Recommendations

Introduced in V4

WHAT IT DOES

Based on income, spending breakdown, and active savings goals, Gemma generates specific data-grounded recommendations. Example: 'Your Food spending is ■2,400 above your 3-month average. Reducing to average would reach your Laptop goal 2 months sooner.' Max 3 shown.

WHY IT MATTERS

Generic advice ('spend less') is useless. Personalised recommendations with specific category names, amounts, and timeline impacts are actionable and motivating.

TECHNICAL DETAILS

- Recommendations generated weekly or on user request
- Each references a specific category and rupee amount
- Goal timeline impact shown: 'saves N months'
- User can mark a recommendation as 'following' to track progress
- Maximum 3 at a time to avoid overwhelm
- Only recommends cuts to discretionary categories — never fixed expenses like rent

V5

Polished Release

The definitive release — every detail refined, accessibility built in, ready for the Play Store.

V5 is not about new features. It is about perfecting everything built in V1 through V4. Every screen, animation, and edge case is reviewed and refined. Accessibility is fully implemented. Onboarding guides new users thoughtfully. Local encrypted backup provides data durability. Performance tuned for years of use. V5 is the Play Store submission.

DESIGN & UX

Full UI / UX Refinement

Introduced in V5

◆ **UX INSIGHT** · Research shows a functionally correct app that feels rough is less trusted and less used. Polish communicates care. For a finance app where trust is paramount, this investment matters more than in almost any other category.

WHAT IT DOES

Comprehensive design pass over every screen: micro-animations, smooth shared-element transitions, illustrated empty states, standardised spacing grid, and typography audit. The app should feel as polished as any top-tier commercial finance application.

WHY IT MATTERS

Every interaction in MoneyTracker is about trusting the app with your financial life. Visual polish directly signals that the developer takes their experience and data seriously.

TECHNICAL DETAILS

- Shared element transitions between list items and detail screens
- MPAndroidChart load animations on every chart
- Illustrated empty states on all major screens
- Standardised 8dp spacing grid applied across all layouts
- Typography audit: heading hierarchy, line heights, contrast ratios
- Haptic feedback on all confirmations, deletions, and goal completions

DESIGN & UX

Accessibility (A11y)

Introduced in V5

WHAT IT DOES

Full Android accessibility compliance: TalkBack content descriptions on every interactive element, WCAG AA contrast ratios on all text, 48dp minimum touch targets, and dynamic text size scaling tested at 85%, 100%, 130%, and 150% system font scale.

WHY IT MATTERS

Financial management should be accessible to everyone, including users with visual impairments or motor difficulties. Accessibility is not optional.

TECHNICAL DETAILS

- TalkBack descriptions on all icons, charts, and interactive elements
- WCAG AA: minimum 4.5:1 contrast on all body text
- Touch targets: all tappable elements $\geq 48dp \times 48dp$
- Layouts tested at 4 font scale levels
- Focus navigation order correct for keyboard and switch access users

DESIGN & UX

Onboarding Flow

Introduced in V5

◆ **UX INSIGHT** · YNAB research shows getting users to their first logged transaction within 2 minutes of install is the single most important retention metric for any finance app. The privacy promise step builds trust before the user even uses a feature.

WHAT IT DOES

Friendly illustrated multi-step first-launch experience: (1) Welcome + privacy promise, (2) Set base currency, (3) Customise initial categories, (4) Set first budget (skippable), (5) Set up app lock PIN, (6) Brief feature tour. Completable in under two minutes.

WHY IT MATTERS

First impressions determine whether a user returns. Onboarding that explains the privacy model and gets users set up correctly increases day-7 retention far more than dropping users onto a blank dashboard.

TECHNICAL DETAILS

- Step progress indicator throughout (step N of 6)
- Skip available at every step except PIN setup
- Privacy promise step: no internet, no servers, no ads — verifiable
- Illustrated screens using custom SVG or vector illustrations
- Skipped steps accessible any time from Settings

DATA

Local Encrypted Backup & Restore

Introduced in V5

WHAT IT DOES

Export AES-256 encrypted backup of the entire database — all transactions, categories, budgets, goals, settings — as a single .mtbackup file to a user-chosen local folder via Storage Access Framework. On a new device: install → Restore → select file → enter PIN → full history restored.

WHY IT MATTERS

Without backup, all data is lost if the device is lost, damaged, or reset. A local encrypted backup provides data durability without cloud dependency.

TECHNICAL DETAILS

- File format: .mtbackup (AES-256 encrypted SQLite export)
- Same key derivation as main database (PBKDF2 + user PIN)
- Typically under 5 MB even for years of transaction history
- Saved via SAF — user chooses destination folder
- Full database replacement — partial restore not supported

CORE

Performance Tuning

Introduced in V5

WHAT IT DOES

Systematic performance audit: Room Database query plans reviewed and optimised, N+1 query patterns eliminated, Paging 3 library applied to the transaction list (loads 30 items at a time), battery usage profiled with Android Battery Historian, cold start targeted under 800 ms.

WHY IT MATTERS

An app that slows as data grows, or drains battery, will be uninstalled. Performance is a baseline expectation for an app used daily over multiple years.

TECHNICAL DETAILS

- EXPLAIN QUERY PLAN used on all Room queries during audit
- Paging 3: transaction list loads 30 items at a time
- Cold start target: under 800 ms
- Memory profiling: zero leaks under standard usage
- WorkManager jobs deferred to charging periods where possible

CORE

Comprehensive Settings Screen

Introduced in V5

WHAT IT DOES

Fully organised settings grouped by section: General (currency, language, theme), Security (PIN change, biometrics toggle, lock timeout), Notifications (bill reminders, budget alerts, goal nudges), Data Management (CSV export, backup, restore, clear all), and About (version, open-source licences, privacy policy).

WHY IT MATTERS

Settings are where users go to fix, change, or understand something. A well-organised settings screen builds trust and reduces confusion.

TECHNICAL DETAILS

- Grouped with section headers and dividers
- Destructive actions (clear data) require confirmation dialog
- Privacy policy displayed in-app — no external link, no internet
- Open-source licence screen listing all bundled dependencies
- Version number and build date shown for support purposes

CORE

Play Store Release Preparation

Introduced in V5

WHAT IT DOES

All assets required for a professional Play Store listing: polished phone and tablet screenshots, feature graphic, a short description focused on privacy, a full description, content rating, target SDK compliance, 64-bit binary support, Play App Signing enrolment.

WHY IT MATTERS

The Play Store listing is the user's first impression before installing. A professional listing with clear screenshots and a compelling privacy-focused description meaningfully improves install conversion.

TECHNICAL DETAILS

- Screenshots: min 2, max 8, phone and 7-inch tablet breakpoints
- Feature graphic: 1024 x 500 px

- Short description (80 chars): privacy + local-first message
 - Content rating: Everyone
 - Target SDK: latest stable at V5 development time
 - Privacy policy at a stable URL (GitHub Pages acceptable)
-

Complete Feature Summary

All 44 features across all six phases — with category, version, internet requirement, and Android permissions needed.

Feature	Category	Version	Internet?	Permission
Expense & Income Entry	CORE	V1	No	None
Custom Categories	CORE	V1	No	None
SQLCipher Encrypted Database	SECURITY	V1	No	None
No Internet Permission	PRIVACY	V1	No	None
Monthly Dashboard	DESIGN & UX	V1	No	None
Reports — Charts	ANALYTICS	V1	No	None
Dark & Light Theme	DESIGN & UX	V1	No	None
PIN & Biometric App Lock	SECURITY	V1	No	USE_BIOMETRIC
Category Budgets	CORE	V2	No	None
Rollover Budgets	CORE	V2	No	None
Savings Goals	CORE	V2	No	None
Bill & Recurring Reminders	NOTIFICATIONS	V2	No	None
Calendar View	ANALYTICS	V2	No	None
CSV Data Export	DATA	V2	No	WRITE_EXTERNAL
Transaction Search & Filter	CORE	V2	No	None
Home Screen Widget	DESIGN & UX	V2	No	None
Safe-to-Spend Number	DESIGN & UX	V♦	No	None
'Since Last Opened' Summary	DESIGN & UX	V♦	No	None
Pre-Committed Bills on Budget	DESIGN & UX	V♦	No	None
Bulk Transaction Actions	CORE	V♦	No	None
Monthly Running Totals	DESIGN & UX	V♦	No	None
At-Risk Category Highlights	NOTIFICATIONS	V♦	No	None
Swipe Gestures on Tiles	DESIGN & UX	V♦	No	None
Subcategories	CORE	V♦	No	None
SMS Transaction Auto-Reading	CORE	V3	No	READ_SMS
Cash Flow Projections	ANALYTICS	V3	No	None
Debt Payoff Planner	ANALYTICS	V3	No	None
Net Worth Tracker (Manual)	ANALYTICS	V3	No	None
Multi-Currency Support	DATA	V3	No	None
Spending Pattern Analysis	ANALYTICS	V3	No	None
Advanced Reports	ANALYTICS	V3	No	None

Feature	Category	Version	Internet?	Permission
MediaPipe / Gemma AI	AI	V4	No*	None
AI Spending Forecasts	AI	V4	No	None
Natural Language Queries	AI	V4	No	None
Anomaly Detection	AI	V4	No	None
Smart Category Suggestions	AI	V4	No	None
AI Savings Recommendations	AI	V4	No	None
Full UI / UX Refinement	DESIGN & UX	V5	No	None
Accessibility (A11y)	DESIGN & UX	V5	No	None
Onboarding Flow	DESIGN & UX	V5	No	None
Local Encrypted Backup	DATA	V5	No	WRITE_EXTERNAL
Performance Tuning	CORE	V5	No	None
Comprehensive Settings	CORE	V5	No	None
Play Store Preparation	CORE	V5	No	None

* Gemma model weights are downloaded once at app update time. All subsequent AI inference is fully offline. INTERNET permission is never added.